



# Matteo Riva

07.02.1994



Lugano (CH)



Permit C EU/AELS



teo.riva



matteoriva@me.com



+41 (0) 782174193



ascarrambad

Experienced computer vision and software engineer. Passionate about precision, simplicity, and well-structured solutions. Eager to contribute to a technical and creative team to develop cutting-edge products.

## Work Experience (a)

### Artificialy

Lugano, Switzerland

#### Computer Vision and Software Engineer

02.2026 — ongoing

- Developed an api-hub + webportal providing access to all company products and services, used for client demos, internal usage, and commercial api access  
`python, java, gitlab ci/cd, docker, uv, atlas, fastapi, traefik, keycloak, qdrant, postgres, valkey, garage/seaweedfs, openai-compliant apis`
- Refactored a PoC project into production-ready system running on Jetson hardware for real-time inspection in a pavents-tile production facility  
`python, c, docker, aravissrc+deepstream, zeromq, opencv, numpy`
- Contributed to the design and development of a microphone and VAD system evaluation pipeline to improve audio segmentation for an interactive agent  
`python, javad, silero, mlflow`

### Delvitech

Mendrisio, Switzerland

#### Senior Computer Vision and Software Engineer

12.2024 — 01.2026

#### Computer Vision and Software Engineer

06.2023 — 12.2024

- Organization of sprints and tasks among cross-functional teams
- Collection of product requirements in collaboration with the PO and translation into a technical specifications and functional requirements
- Design of the overall system architecture, with a focus on vision modules and system behavior
- Designed and developed a distributed PatchCore **anomaly detector implementation in a real-time prod environment**; whole pipeline runs in ~90 ms  
`python, java, grpc, torch, faiss, scipy, onnx-runtime, opencv, numpy`
- Designed and developed an image preprocessing barcode reader pipeline, supporting 1D and 2D codes printed, laser-etched, heavily damaged, etc.; whole routine runs in ~100 ms  
`java, opencv, zxing, libdmtx`
- Several JNIs to integrate C++ libraries with the Java runtime  
`c, c++, java, libdmtx, zxing-cpp`
- Designed and developed an optical head simulator using a physics-based rendering system to create synthetic imagery  
`python, mitsuba, pybullet, scipy, opencv, numpy`
- Trained and deployed several production vision DL models under **data availability, inference time, and memory constraints**  
`python, opencv, imgaug, torch, onnx, tensorrt`
- Several projects involving **classical computer vision** techniques, including **phase shift profilometry for 3D reconstruction**  
`python, c++, numba, cupy, scipy, numpy, opencv`

### IDSIA — Dalle Molle Institute for AI

Lugano, Switzerland

Funds by Innosuisse & SNSF

#### Machine Learning Researcher

09.2020 — 10.2022

#### Teaching Assistant

09.2020 — 10.2022

- Graph reconstruction** and multivariate **time series analysis and forecasting**
  - day-ahead prediction of photovoltaic energy production**
  - rainfall-runoff model to infer the likelihood of floods in a given river**
- Research focus on spatiotemporal data processing using **graph representations** and **Graph Neural Networks**  
`python, torch, torch-geometric, numpy, pandas, geopandas, mongodb`
- Prepared and graded homework assignments, midterm projects, and written exams for the **Databases (M.Sc.)** and **Machine Learning (B.Sc.)** courses
- Prepared and taught the practical classes and labs for both courses  
`mysql, mongodb`



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## Work Experience (b)

### Machine Learning Intern

03.2019 — 10.2019

- Developed a **lipreading system for speaker-independent word recognition** using **Recurrent Neural Networks (LSTM)** and **CNNs**

*python, sacred, tensorflow 1&2, numpy*

### Monkey Cube & ExMachina

Milan, Italy

### iOS Software Engineer

2014 — 2016

- Designed and developed backend and frontend of three native iOS apps written in Swift from scratch, including widgets and share extensions
- Porting of old code base from Objective-C to Swift

*swift, objective-c, ios frameworks, cocoapods, 3rd-party rest apis*

## Personal Projects

### Linear motion-controller utility

- Developed a small utility to remotely connect via EtherCAT to an HIWIN motion controller to perform basic axis-independent movements using low-level position-control APIs

### Several opensource JNIs for 2D code decoding

- Developed JNIs for libquirc and libwechat-zxing-cpp (part of OpenCV-contrib) to enable 2D QR and Datamatrix decoding with high performance C/C++ libraries, handy if you're codebase is in Java

### Fast NN Prototyping Framework

- Scalable software framework exploiting TensorFlow** for neural network building using text strings
- Proved **effective for scaling from simple toy models to complex deep learning architectures**

### Hand detection and hand-gesture classification for IoT

- Controls the specified IoT devices over the network**, allowing users to turn them on/off or adjust their settings with hand gestures

### Several iOS apps, some recognized as "Best of the week"

- Remote mouse control Wii-style, public transports, music player, blog and podcast, tv-shows scheduler

## Practical Skills

### Vision models

- YOLO, UNet, ResNet, Siamese networks

### 3D Reconstruction & Generation

- Phase-Shift Profilometry (PSP), NeRFs
- open3d, pydraco

### Forecasting models

- RNNs, Transformers, Temporal Convolutional Networks, Graph Convolutional Networks

### Miscellaneous

- pyqt, qt-designer, matplotlib, bokeh, plotly
- git, docker, docker-compose, nvidia-triton
- shell, cmake, vcpkg

## Language Skills

- native Italian
- fluent English — IELTS certificate
- A2/B1-level German

## Soft Skills

- Collaboration in multicultural environments
- People-coordination, organizational skills
- Quick learner with strong problem-solving skills
- Attention to detail and focus on quality
- High sensibility to overall product functionality, capabilities, and software coherence

## Education

### ETH-CSCS Summer School

- GPU & ARM architectures** fundamentals - **CUDA, OpenACC**, and **Kokkos**

2022

### M.Sc. in Informatics and M.Sc. in Computer Science

10.2017 – 04.2020

- Università della Svizzera italiana (CH) — Università degli Studi di Milano-Bicocca (IT)**  
Thesis: "Speaker-independent Lipreading with Neural Networks Exploiting Motion Dynamics"

### B.Sc. in Computer Science

09.2014 – 10.2017

- Università degli Studi di Milano-Bicocca (IT)**  
Thesis: "CareLine: iterative design and development of an iOS app guided by a usability study"